

DSA / Algorithms (Abdul Bari) - Day 2

45-day plan mapped to the Algorithms playlist

Writing & Analysing Algorithms

How to write an algorithm and the frequency-count method for analysis.

Watch these Abdul Bari videos

- > 1.3 How Write and Analyze Algorithm
- > 1.4 Frequency Count Method

Key concepts

- Count how many times each statement runs (frequency count).
- Sum the counts to get total operations as a function of n .
- This gives the time function before Big-O simplification.

Key idea

A loop 'for $i=1..n$ ' runs $n+1$ times (including the failing test).

Tasks (1.5h)

- ☐ Watch both videos
- ☐ Do a frequency count of a loop
- ☐ Express the count as $f(n)$

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0